Candidate Number ____________________ Fraction ____________________

## **2017** International Biology Olympiad National Team Selection Camp Practical Test Questions

# First **Trial** Session **B**

### Part **II :** Restriction Enzyme Site Detection of Plastid Composition

All the equipment and reagents needed for the experiment are on the table. Please check them according to the list below.

If anything is missing, please raise your hand and let the judges know. After the experiment, please clean and dry all used equipment.

Clean and neatly arranged.

Experimental equipment:

| Equipment Quantity: Microdispenser p20 1 piece; Sterile | | Pharmaceuticals and materials | Quantity (µl) |
|---|---|---|---|
| water p20; Microdispenser pipette tips 30 pieces | | | 1 tube (100) |
| | | | |
| Place up to 4 microcentrifuge tubes containing the medicine on ice. | | | |
| 1 marker pen, 1 tube of 5X (multiplied) restriction enzyme reaction buffer (15 tubes) | | | |
| Timer 1, EcoRI restriction enzyme 1 tube (10) | | | |
| 1 tube of glue, 1 tube of XhoI restriction enzyme (10 tubes) | | | |
| Mini centrifuge, 1 plasmid A, 1 tube (4) | | | |
| Numbered centrifuge tubes, water bath float, 1 mass B, tube (4) | | | |
| Numbered petri dishes (for holding film), 1 plastid C, 1 tube (4) | | | |
| Electrophoresis tank 1 tube (15) | 1 DNA sample with dye (labeled: Dye) Shared ruler for DNA | | |
| Water bath (37°C) 1 tube (20) | (labeled: M) Shared | | |
| DNA film imaging system | | | |
| | | | |

Please note:

1. Please confirm that your candidate number is correct; if it is incorrect, please raise your hand and ask the teaching assistant for assistance.

2. Once the medicines and equipment on the table are used up, they will not be replenished.

3. Please use public instruments in accordance with the instructions.

4. This exam paper (including the cover and the question paper) consists of **6** pages and must be returned in its entirety when submitting the exam.

5. The allotted time is **90** minutes. Please answer the questions on this paper. You may write your answers on the back of the questions, but please indicate the question number.

## Part II : Restriction Enzyme Site Detection of Plastid Composition (50 points)

Background information:

Cells often regulate the expression of related genes based on the nutrient status in the environment. In budding yeast, this is related to galactose...

The regulation of galactose metabolism-related genes is a well-studied example (Figure 1). This occurs when glucose is absent in the environment.

(glucose), but in the presence of galactose, the transcription factor Gal4 is activated, which increases the expression of related genes (GAL genes), inhibiting...

Proteins involved in galactose transport and metabolism, including Gal2, Gal1, Gal7, and Gal10, were created . Gal2...

Galactose from outside the body is transported to the cytoplasm, where Gal1 converts the galactose in the cytoplasm into galactose-1P.

Gal7 and Gal10 are involved in the conversion of galactose-1P to glucose-1P. The regulation of Gal4 transcription factor activity, however, is...

It is the key component of the entire system; without galactose, Gal4 attaches to the regulatory region of the GAL gene promoter.

(UASGAL), but its activity is inhibited by the inhibitory protein Gal80; when galactose in the environment uses a small amount of pre-existing protein...

Once Gal2 is transported into the cell, another regulatory protein, Gal3, binds to galactose and ATP and then enters the nucleus.

Internally, it eliminates Gal80 and activates Gal4, thereby triggering a large-scale expression of GAL genes.

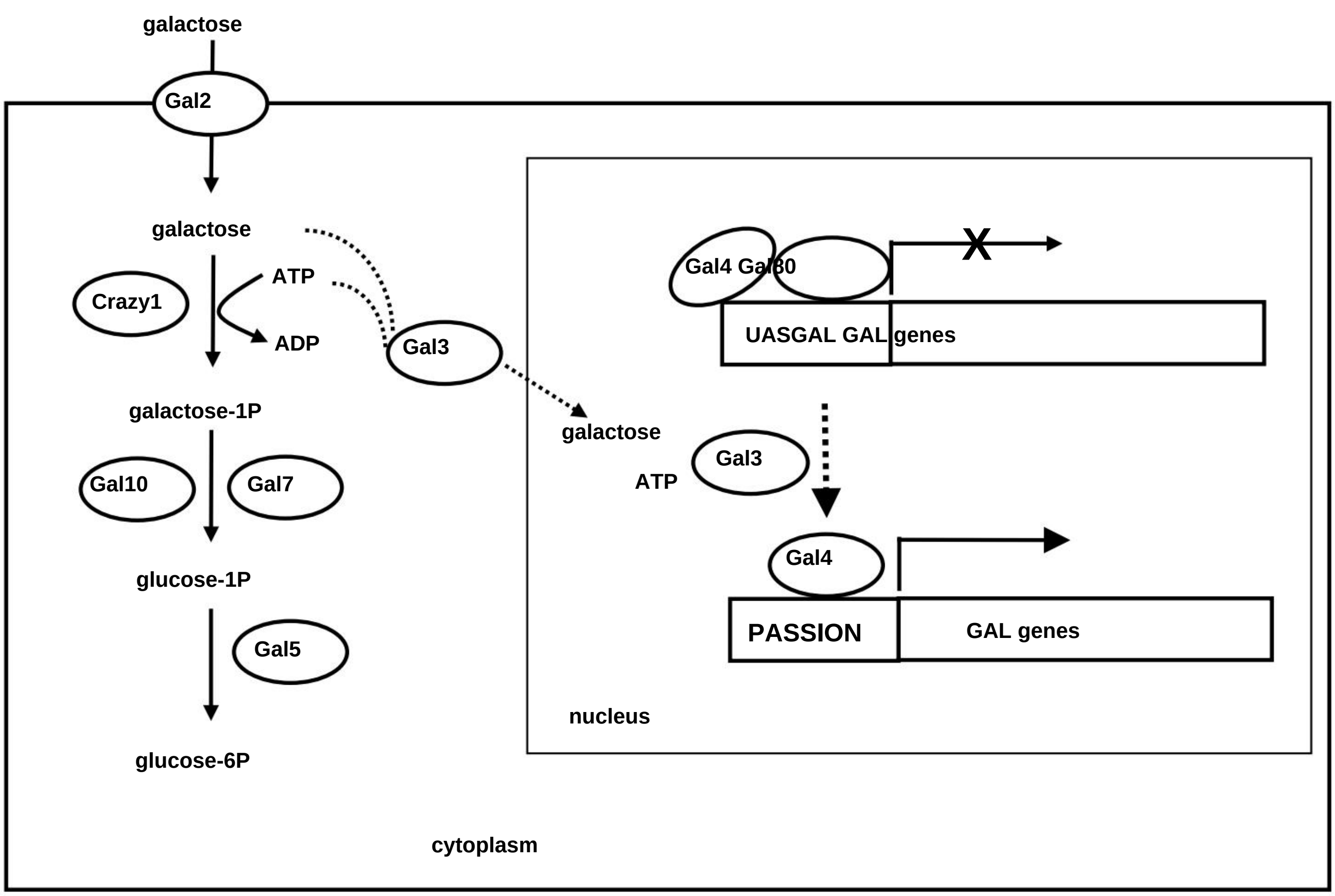



Figure 1. Metabolic pathways and related gene regulatory mechanisms of galactose in yeast.

Since the expression of genes located behind UASGAL can be regulated simply by adding or washing galactose , this GAL system...

It is often used as a tool for ectopic expression of specific genes; simply insert the gene you want to study into...

After UASGAL, its performance can be artificially regulated. The yeast N80 protein is normally only activated during meiosis.

In his study, graduate student Meng Yu used the GAL system to investigate whether the N80 protein has an effect on mitosis. He constructed a model with...

Two hours after adding a small amount of galactose to the culture medium, the UASGAL-N80 strain showed a significant increase in N80 protein levels.

The N80 was initially present inside the cells, but after two hours, only trace amounts remained. Meng Yu wanted to prolong the N80-induced expression.

He was currently limited in time, but due to experimental constraints, he could not increase the amount of galactose he used, which greatly troubled him. Just then...

Professor Hopper, an expert in GAL systems, visited at one time and suggested that he could use one of the proteins in Figure **1.**

Knockout of the coding gene might increase expression time, but the regulation of GAL can still be achieved through the addition and washing of galactose.

The system's on/off switch. Following Professor Hopper's suggestion, Meng Yu first used PCR to analyze this gene (which we'll temporarily represent as the X gene).

Gene X was selected and colonized onto plastids to obtain plastid T789 (Figure 2), and a portion of the coding region of Gene X on plastid T789 was moved.

In addition, a marker gene is inserted into the vacant position after removal for identification in subsequent experiments.

The new plasmid after recombination is plasmid T790 (Figure **3).**

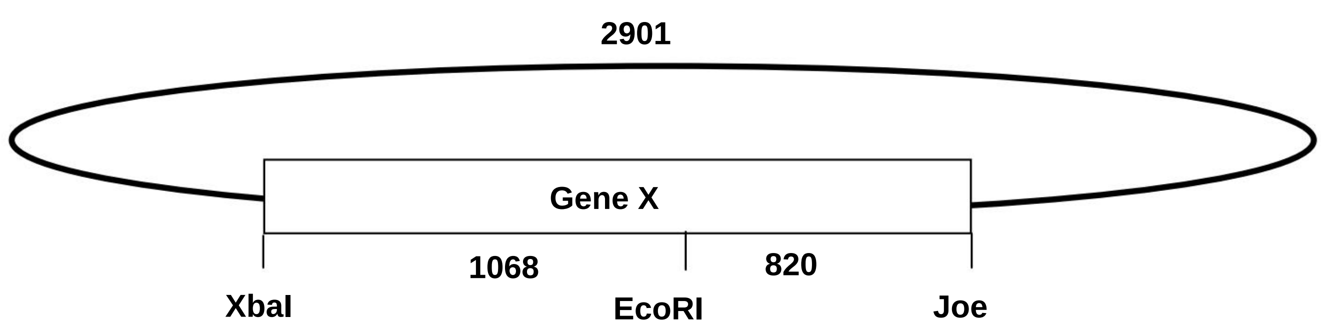



Figure 2. Restriction enzyme sites associated with plasmid T789 and the distances between these sites (unit: bp)

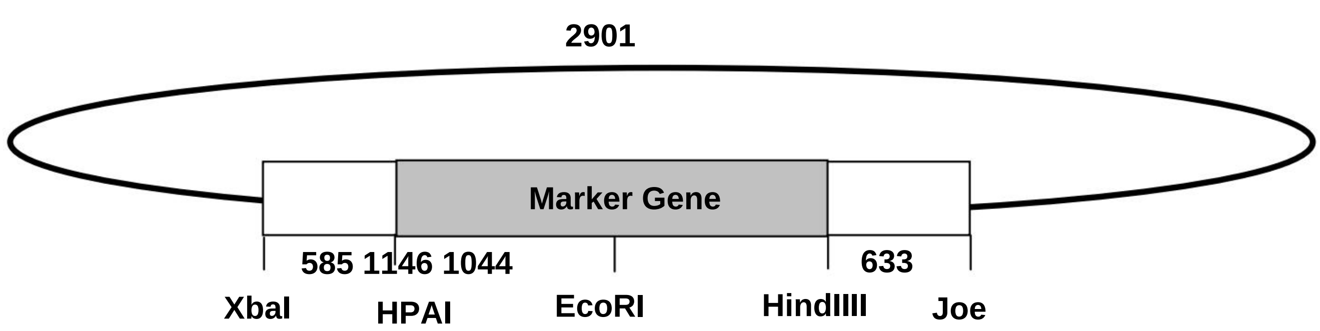



Figure 3. Restriction enzyme sites associated with plasmid T790 and the distances between these sites (unit: bp)

Your experimental task is to use restriction enzyme site detection to identify three plasmids (A, B, and C) in the material, and determine which one is the plasmid.

Which is T789? Which is T790? It is recommended to perform the experimental procedures first, and then answer the question in the meantime.

**Q1.** In Meng Yu's original GAL system, what could be the reason for the decrease in the expression level of N80 protein? (8 points)

**Q2** In order to control the switching of the GAL system through the addition and washing of galactose, and to prolong the period of time after adding galactose.

Regarding the timing of N80 gene expression, which gene should Professor Hopper recommend removing, and why? (8 points)

Experimental steps:

**Q3.** Based on Figures 2 and 3, calculate the yield of plasmids T789 and T790 after simultaneous cleavage with restriction enzymes EcoRI and XhoI.

What are the sizes of the segments? (4 points)

T789:

T790:

**Q4** simultaneously cleaves plasmids A, B, and C with restriction enzymes EcoRI and XhoI, respectively. In a single plasmid cleavage reaction, two restriction enzymes...

Each enzyme was used at a concentration of 4 units, while the concentrations of EcoRI and XhoI in the experimental materials were both 2 units/μl. Based on these conditions,

Fill in the restriction enzyme cleavage reaction design (preparation amount for single plastid cleavage) in the table below. During the restriction enzyme reaction, the reaction...

The mixture should contain a 1X (times) concentration of restriction enzyme reaction buffer. (4 points)

| Formulation ingredients | plastid DNA | 5X (times) concentration restriction enzyme reaction buffer | EcoRI | sterile water | XhoI | Total Volume |
|---|---|---|---|---|---|---|
| Usage (μl) | 2 | | | | | 15 |

1. Prepare a mixture sufficient for the reactions of three restriction enzymes according to the design in Q4: Take an empty microcentrifuge tube and label it A with a marker.

   Following the order of sterile water, 5X (fold) concentration reaction buffer, restriction enzyme EcoRI, and XhoI, add each via microdispenser.

   Add three times the volume of solution listed in Table Q4 to this microcentrifuge tube. When aspirating different solutions, use a clean, unused pipette tip and a micropipette.

   Volume dispenser for suction and discharge mixing.

2. Take two more empty microcentrifuge tubes, label them B and C respectively, and use a microdispenser to mix the restriction enzyme reaction in tube A.

   Take 13 μl of each solution and add them to tubes B and C respectively.

3. Using a microdispensing device, aspirate 2 μl of DNA from plasmid A, plasmid B, and plasmid C respectively, and add the corresponding restriction enzyme reaction mixture.

   The mixture should be aspirated and mixed using a microdispenser in tubes A, B, and C. Each time a sample is aspirated, use a clean, unused syringe.

   The tip of the straw.

4. Centrifuge the mixture in each tube to the bottom of the tube using a mini centrifuge. When centrifuging, please make sure to place the centrifuge tubes evenly in the centrifuge.

5. Place the centrifuge tubes containing the reaction mixture on a numbered water bath float and incubate in a 37°C water bath for 40 minutes (using a timer).

   (Timed), use the waiting time to answer questions or complete the practical questions in Part I.

6. When the 40-minute reaction time is over, remove your sample from the water bath.

7. Add 3 μl of DNA dye to each reaction sample using a microdispenser, and mix by aspiration and dispensing using the dispenser.

8. Using a microdispenser, dispense the reaction sample containing the added dye and 18 µl of ruler-marked DNA separately, according to the ruler-marked DNA...

Inject the sample into the sample wells of the electrophoresis film in the order of tubes A, B, and C, from left to right. Add the sample gently.

Do not overflow the groove.

9. Raise your hand to signal the lab assistant, and follow their instructions to connect the power supply and perform electrophoresis for 25 minutes. Please be careful not to touch it.

Contact electrode.

10. After the electrophoresis begins, the teaching assistant will provide a standard electrophoresis result image. Paste the image in the blank space of Q5 and answer the questions based on the image.

11. After electrophoresis is complete, turn off the power, carefully remove the intact film, place it on a numbered culture dish, and take it to the film imaging center.

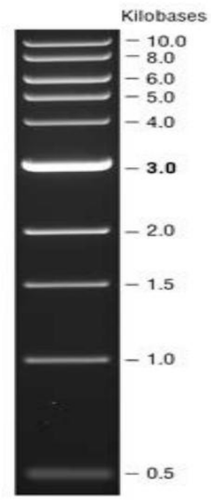
After handing the experiment to the teaching assistant, return directly to your own lab table. The electrophoresis results will be graded by the evaluating teacher.

Experimental results:

**Q5.** Based on the standard electrophoresis results and the standard diagram of DNA separation pattern (bottom left, unit: kb), determine plasmids A, B, and C.

C can be either T789 or T790, or neither. (6 points)

Please paste the standard electrophoresis result image here ÿ




Plasmid A: plastid B: plastid C:

**Q6** The electrophoresis results will be scored by the evaluating teacher based on the electrophoresis film images (15 points).

**Q7.** If N80 is to be able to maintain its expression without being induced by galactose, which gene should be removed? (5 points)

Candidate Number ______________________ **Score**_____________________

2017 International Biology Olympiad National Team Selection Camp Practical Test Questions

First Trial Session **A**

Biochemistry Experiment

Note: Operation time is limited, please make good use of the time waiting for the experiment to take effect.

ÿ All the necessary equipment and reagents for the experiment are on the table. Please check them against the list below. If anything is missing, please raise your hand to let me know.

Dear examiners, after the experiment, please clean and neatly store all used equipment.

Experimental equipment and reagents:

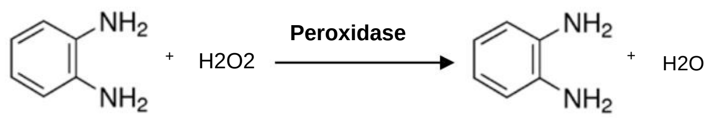
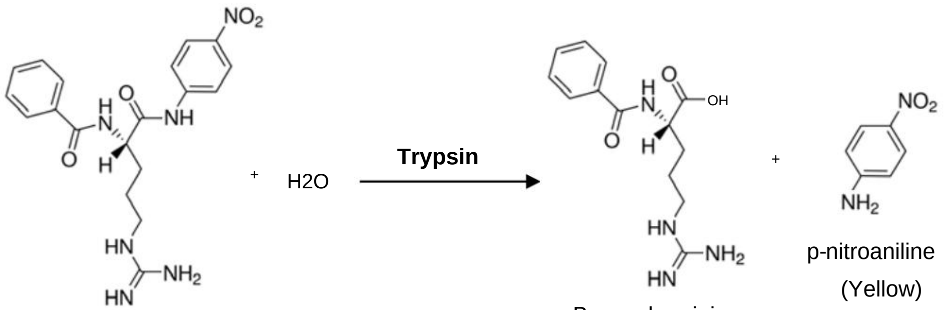
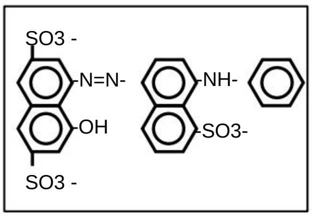
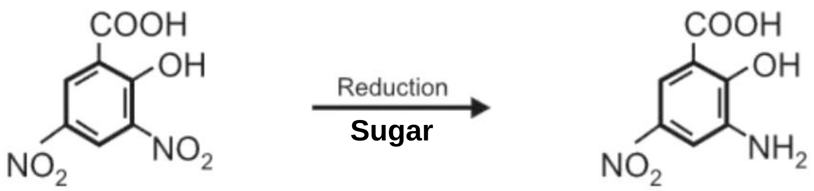
| equipment | quantity | Reagents and experimental | quantity |
|---|---|---|---|
| Microdispenser P20 (shared with | materials: 1 unknown vial, 1 | | 120 ÿL |
| Practice B) | unknown vial, 1 unknown vial, 1 | | 120 ÿL |
| Microdispenser P200 | unknown vial, 1 unknown vial, 1 | | 120 ÿL |
| Microdispenser P1000 | box of unknown vials, 1 box of | | 120 ÿL |
| Micropipette tip 20~200 ÿL Micropipette tip 1000 ÿL | unknown vials, 1 box of unknown vials. | | 120 ÿL |
| 96-well micro-volume plate | 1 ÿ Bradford dye | | 1.2 mL |
| 1.5 mL empty microcentrifuge tube | 5 | Benzoyl-arginine p-nitroanilide (BAPNA) in 50 mM Tris, pH 7.8 o- | 0.6 mL |
| Microcentrifuge tube rack | 1 | Phenylenediamine (OPD) + $H_2O_2$ in 50 mM MES, pH 6 | 0.6 mL |
| Heating plate (100ÿ, please be careful when using!!) Shared | | Ammonium molybdate + Malachite green (MG) | 1.2 mL |
| Waste liquid cup | 1 ÿ 3,5-dinitrosalicylic acid (DNS) 1 ÿ 2 N | | 1.2 mL |
| for ballpoint pens | HCl | | 1.2 mL |
| Timer | 1 $H_2O$ 1 | | 1.2 mL |
| explosion- | | | |
| proof clip float | 1 | | |
| (blue) gloves | 1 pair | | |

## Please note:

1. The materials and equipment on the table will not be replenished once they are used up.
2. This exam paper (including the cover and the question paper) has a total of 4 pages and must be returned in its entirety when submitting the exam.
3. This exam includes Practice A and Practice B, with a total time limit of 90 minutes.
4. Please answer the questions on this paper. You may write your answers on the back of the paper, but please indicate the question number.

I. Experiment Topic: Biochemistry Experiment

A teaching assistant has accidentally mixed up five enzymes or reagents needed for a planned biochemistry experiment. These samples are unlabeled and each contains exactly 60 µL. The experiment is scheduled to begin in 30 minutes. What should he do? He knows the samples contain catalase, trypsin, glucose, BSA, and PBS. Please refer to the following experimental principles to help him identify the samples.

| sample | Reaction |
|---|---|
| catalase<br>Peroxidase | <br>OPD (Colorless) (Yellow/Brown yellow) |
| trypsin<br>Trypsin | <br>BAPNA (Colorless) Benzoyl-arginine p-nitroaniline (Yellow) |
| BSA | BSA +  (Bradford dye) will change the color from brown to blue.<br>The higher the protein concentration, the deeper the blue color. |
| PBS | $P_i + (NH_4)_2MoO_4 \xrightarrow{H^+} H_3PMo_{12}O_{40}$<br>$H_3PMo_{12}O_{40}$ (yellow) + $HMG^{2+}$ (malachite green) (yellow, $\lambda_{max}$ 446 nm) $\xrightarrow{H^+}$ $(MG^+)(H_2PMo_{12}O_{40})$ (green, $\lambda_{max}$ 640 nm) + $2H^+$ |
| glucose<br>Glucose | <br>3,5-dinitrosalicylic acid (yellow) 3 amino, 5-nitro salicylic acid (orange-red) |

## II. Experimental Procedure: (10 points)

Experiments 1 through 4 below were all conducted in a 96-well microplate, and the arrangement and procedures of the experiments are as follows:

**1.** The first row in the horizontal direction is for catalase activity analysis.

Take 20 ÿL of each unknown sample and place it into the corresponding 96-well microplate. Then add 100 ÿL of OPD/ $H_2O_2$ matrix solution and react at room temperature for 10 minutes. Then add 10 ÿL of 2N HCl to terminate the reaction.

**2.** The second row in the horizontal direction is used for trypsin activity analysis.

Take 20 ÿL of each unknown sample and place it into the corresponding 96-well microplate. Then add 100 ÿL of BAPNA matrix solution and react at room temperature for 10 minutes. Finally, add 10 ÿL of 2 N HCl to terminate the reaction.

**3. The third row in the horizontal direction is for Bradford dye-binding protein quantification.**

Take 20 ÿL of each unknown sample and place it into the corresponding 96-well microplate, then add 200 ÿL of Bradford colorimetric reagent.

**4. The fourth row in the horizontal direction is the Malachite green phosphate assay.**

Take 20 ÿL of each unknown sample and place it into the corresponding 96-well microplate, then add 200 ÿL of Ammonium molybdate/Malachite green colorant.

The fifth experimental unit below will be conducted first in a microcentrifuge tube. After the experiment is completed, 200 µL of sample will be taken from the centrifuge tube and placed in the fifth horizontal row of the 96-well microplate.

**5. Reduction sugar assay** for quantification of reducing sugars

Take 20 ÿL of each unknown sample and place it into a microcentrifuge tube. Then add 200 ÿL of DNS reaction solution and heat at 100ÿ for five minutes. After the reaction is complete, move the microcentrifuge tube to room temperature and cool for 3 minutes. Then add 200 ÿL of $H_2O$ to each microcentrifuge tube, mix well, and then take out 200 ÿL from each tube and place it in the fifth horizontal row of a 96-well microcentrifuge dish.

III. Experimental Results and Question Answering: Only those experiments that are completed and successfully completed will be scored. No points will be awarded if there are no experimental results that can be used to infer or corroborate the identity of the unknown sample.

1. Please fill in the experimental results in the blanks of the following 96-well microvolume tray according to the colors shown in your tray. Mark yellow as Y, dark blue as B, dark green as G, and orange-red as R. Do not mark the other colors. (20 points)

X1 X2 X3 X4 X5

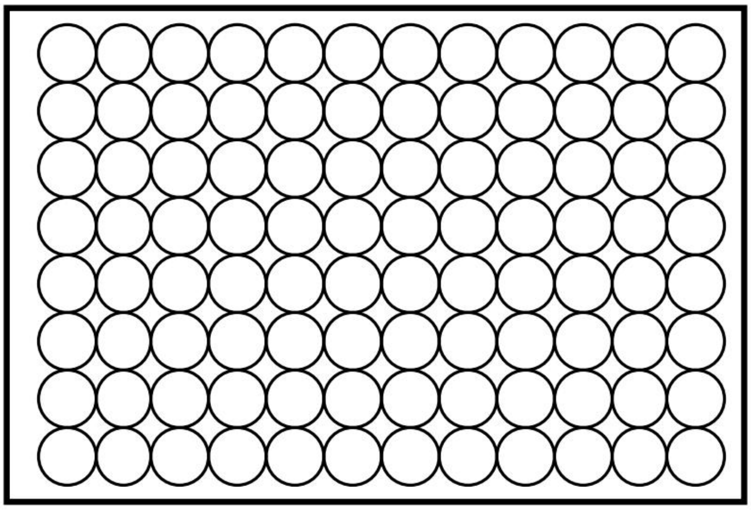
OPD

MEN

Bradford

Malachite

DNS



2. Please write down the identity of the unknown sample (10 points)

| Unknown samples | Real identity |
|---|---|
| X1 | |
| X2 | |
| X3 | |
| X4 | |
| X5 | |

3. If sucrose is used for the fifth experimental unit, what will the color be after the reaction? And why? (6 points)

4. Why can trypsin hydrolyze BAPNA? (4 points)